

Exploration of Automatic Speech Recognition for Deaf and Hard of Hearing Students in Higher Education Classes

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ABSTRACT
Automatic speech recognition (ASR) programs that generate real-time speech-to-text captions can be provided as supplemental access technologies for deaf and hard of hearing (DHH) students in higher education classes. As part of a pilot program, we implemented ASR as a supplemental access service in biology, statistics, and other courses at our university. To identify the benefits and limitations of ASR as an access technology, we surveyed 26 DHH students and interviewed 8 of these students about their experiences with ASR in their mainstream classes. Participants believed that ASR was beneficial despite the errors that ASR continued to generate; however, the accuracy and readability of ASR need to improve so that students can better access spoken information through ASR. This paper reviews points for researchers to consider when designing and providing ASR as a supplemental access service in educational settings.

Author Keywords
Human-centered computing~Accessibility technologies; Human-centered computing~Accessibility design and evaluation methods; Human-centered computing~Empirical studies in accessibility

ACM Classification Keywords
Deaf and Hard of Hearing; Automatic Speech Recognition; Real-Time Captions; Feedback.

1. INTRODUCTION
Real-time speech-to-text systems, also known as automatic speech recognition (ASR) engines that generate captions, can be provided as access technologies for deaf and hard of hearing (DHH) students in mainstream classrooms. Other captioning services, including C-Print and CART (Communication Access Real-Time Translation), are provided by human captionists who may not always be available for hire at different institutions. ASR generated captions provide students with an additional means of accessing spoken language, especially when other access services are not readily available. While ASR is becoming more widely used across different platforms in more settings, researchers need to optimize the readability and accuracy of automatically generated captions for students.

As part of a pilot program, we continue to implement ASR as a supplemental access service in biology, statistics, and other courses at our university. Automatically generated captions appear next to course slides in combination with other access services in the classroom, including sign language interpreters, as shown in Figure 1.

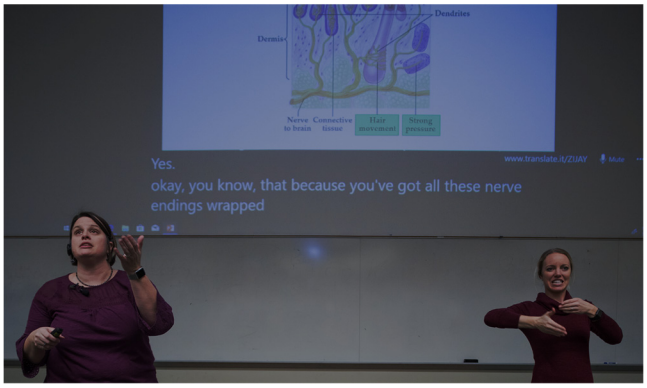


Figure 1. ASR shown below PowerPoint in a biology course

In order to understand students' responses to reading ASR generated captions in their biology and statistics classes, we surveyed 26 DHH participants and interviewed 8 of these 26 participants. This qualitative study is unique in assessing DHH students' experiences with using ASR as a supplemental access service in classrooms with sign language interpreters and human captionists. Our findings suggest that while DHH students seem to benefit from ASR engines that supplement other access services, students are concerned with the accuracy and readability of ASR. Their perspectives inform future research studies that should aim to improve the readability and accuracy of ASR before providing ASR as a primary access service for students.

1.1 ASR in the Classroom
As ASR becomes more viable as an access service, researchers have studied the use of real-time speech transcription in various classroom settings for a variety of

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learners [1, 8, 15, 23]. These studies reveal the benefits of ASR for students and particularly for DHH students who receive access to real-time spoken information.

While ASR can be beneficial, the residual rates of inaccuracy and latency, or delay, of automatically generated captions indicate that ASR still needs to be improved before being used as the primary access service in the classroom. Researchers have been working for years on human-computer collaboration to raise the accuracy rates of ASR, particularly through human supervision or editors [11, 16, 17, 21, 22] and through customizing ASR for speech variability [2]. Fujii et al. [10] presented a method for improving the readability of ASR by using an algorithm to help edit transcripts in order to support readers' understanding of course material. Their review of readability included disfluencies as well as unclear sentence boundaries, punctuation marks, and missed or repeated words. In addition to these points, our interviews with participants revealed that students' ability to read and understand ASR generated captions can also be influenced by the rate at which words appear on screen in real time, how words are spelled, and the accuracy of course-specific keywords.

Other researchers have made progress with predicting and improving the Word Error Rate (WER) of ASR so that ASR can more accurately recognize and transcribe spoken words [4, 5, 19]. While improving the accuracy of ASR is crucial, WER has been critiqued for not correlating with or predicting the performance scores of participants, and researchers have tested other metrics for predicting the usability of ASR [7, 12] or supplemented WER with Semantic Error Rate [13].

Qualitative studies can compensate for the limitations of WER metrics by investigating human subjects' responses to ASR [20]. Kawas et al. [14] responded to the scarcity of qualitative studies on ASR and developed design requirements and features for human-based and machine-based captioning services. Among other points, the authors' design requirements called for accurate real-time captions with minimal lag that could be read and understood easily. They also mentioned key features, including using dictionaries to increase captioning accuracy for different subjects and providing guidance to instructors. While they reviewed a range of human and automatically generated captioning services, our current study provides additional implications on improving ASR as an access service.

In a larger study on ASR that included qualitative methods, Berke et al. [3] found that accuracy was one of the most important concerns for DHH users. In addition, DHH users were concerned about the reliability of the technology and the potential loss of sign language interpreting services. To provide additional strategies for supporting DHH students' access to speech in real time, we discuss our qualitative study in the following sections.

2. STUDY DESIGN AND METHODS

For the purposes of this specific study, we recruited DHH students who took one of three mainstream courses at our university: General Biology I, Evolutionary Biology, and Introduction to Statistics II. ASR was used in conjunction with other support services in these courses, including sign language interpreters and TypeWell, which is similar to C-Print, a speech-to-text transcription system in which a human captionist condenses and organizes important information so that the meaning, rather than the verbatim, of what was said is conveyed [6]. In contrast to human captionists who condense information, ASR engines can create verbatim transcripts—with exceptions that the program may have been programmed to disregard, such as disfluencies. The rate at which ASR transcribes words on screen may vary depending on the speed and complexity of spoken speech; for instance, words might appear one-by-one relatively quickly or appear in chunks with some latency.

Prior to embedding ASR in these classes, researchers at our institution developed custom language models of ASR through Microsoft Custom Speech Service by feeding the system with course-specific vocabulary and key concepts from glossaries in course textbooks, transcripts from prior semesters created by human captionists, and related sources. Without these custom language models, ASR would often transcribe keywords such as "atom" and "SQL" as "Adam" and "sequel." Such transcription errors can be minimized in the custom language model, and the accuracy rates that students saw in the three courses in our study typically ranged from 85% to 90%.

During each class session, the instructor wears a headset microphone that is directly connected to the ASR service to ensure the best audio stream possible. The automatically generated captions appear in real time in the course's PowerPoint slides on screen in the front of the classroom and on DHH students' personal devices, as indicated in Figure 2.

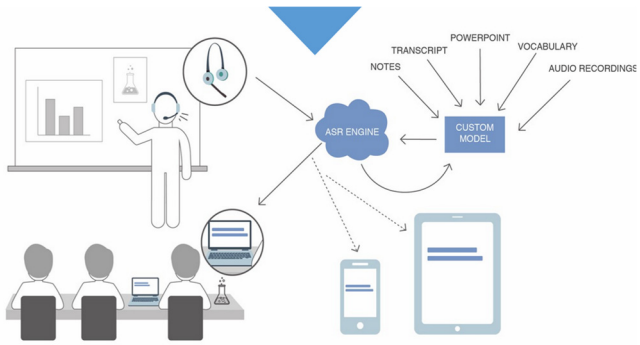


Figure 2. Custom language model of ASR

Instructors briefly mentioned in the courses in our study that course materials were used to improve the accuracy of ASR and notified students that there would still be mistakes, but students did not receive detailed knowledge about the backend. While a custom language model can improve the accuracy of keywords, our qualitative analysis revealed that

punctuation, delay, and other factors can affect the readability of ASR and students' confidence in ASR. In the following sections, we investigate students' perspectives on the benefits and limitations of ASR in order to demonstrate that the readability and accuracy of ASR needs to improve prior to providing ASR as an primary access service.

2.1 Research Goals

We aimed to understand how DHH students responded to ASR in their classes so that we could work towards improving the use of ASR as an access service in mainstream courses. In particular, we were interested in the following research questions:

1. What do students perceive as the benefits and limitations of ASR in class?

2. What is the role of ASR in helping students understand and attend to the instructor and class material?

3. What are students' levels of confidence with using ASR in class?

4. What are students' preferences for the accuracy and latency/timing of ASR in class?

2.2 Procedure

To collect information from a range of students, we contacted DHH students who were enrolled in one of three biology or statistics courses in the previous semester. We asked that they complete an online questionnaire about their experiences with ASR. After 26 participants completed the questionnaire, 8 consented to be interviewed once in person about their experiences with ASR. Questionnaire participants were eligible for a drawing of one of three gift cards and interview participants were compensated for their time. The questionnaire and interviews occurred during the first few weeks of the spring semester so that participants could reflect on and discuss their comprehensive experiences with ASR throughout the fall semester, including any possible changes in their responses to ASR.

A member of the research team who is a nationally certified sign language interpreter conducted separate interviews with all 8 interview participants. Participants were asked to answer 13 questions about their experiences with ASR in the classroom and one question related to specific transcribed passages from class. Four interviewees signed their responses in front of a video camera and four participants spoke their responses into the video camera's microphone. The same member of the research team translated four participants' signed responses to spoken English. All audio files were then processed through Microsoft Video Indexer to generate transcripts. The same researcher then proofread and verified the transcripts for accuracy. We then analyzed the interviews, as detailed in Section 4.

We were able to collect a variety of perspectives from participants with a range of hearing levels, communication preferences, and experiences with different access services. Their differences allowed us to understand how DHH students with a range of learning backgrounds used ASR.

3. QUESTIONNAIRE

A review of the questionnaire participants' responses informed our subsequent analysis of interview participants' detailed responses. Questionnaire participants were asked to answer 28 questions about their communication preferences, experiences using access services, and their responses to ASR in their biology or statistics course.

The 26 DHH students who completed the questionnaire possessed a range of communication preferences. When asked about their communication preferences in face-to-face situations, 9 participants reported feeling most comfortable communicating through American Sign Language or Signed English, 10 reported feeling most comfortable communicating through Spoken English, and 7 reported feeling most comfortable communicating through Simultaneous Communication (signing while speaking).

Participants were asked to self-report their English proficiency, or their skills and ability to read and write in English, on a 5 point scale from Very High Proficiency to Very Low Proficiency. Their responses are depicted in Table 1.

Participants' Responses	English Proficiency
Very High Proficiency	16
High Proficiency	7
Moderate Proficiency	2
Low Proficiency	1
Very Low Proficiency	0

Table 1. Self-reported English proficiency

While participants might not have accurately self-reported their English proficiency, this allowed us to discern information about the perspectives of DHH students with various levels of comfort with spoken and written English.

We aimed to understand their current use of access services and their preferred access services. Participants were asked which one access service they *currently* used for most of their courses at our institution. Their current access services are listed in the first column in Table 2.

Sign language interpreters are often more available to students than other access services, so we asked participants which access service they would *prefer* to use for all of their courses if they could choose any one access service. Their preferences are listed in the second column in Table 2.

The numbers did not change notably between participants' current primary access services and preferred primary access services, suggesting that these students have generally been receiving their preferred access services. The numbers for ASR generated captions impel us to continue to develop

strategies for improving the viability of ASR as an access service.

Participants' Responses	Current Access Service	Preferred Access Service
CART	0	1
None	1	1
FM system	1	1
ASR generated captions	1	4
Notetaker	3	1
C-Print or TypeWell	7	7
Sign language interpreter(s)	13	11

Table 2. Current access services and preferred access services

To explore one of our research questions related to students' levels of confidence with using ASR in class, we asked participants to rate their confidence in the accuracy of ASR in their biology or statistics course on a five-point scale. Figure 3 depicts the number of participants with each level of confidence.

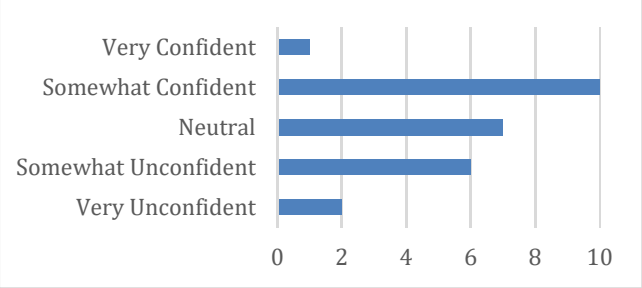


Figure 3. Levels of confidence in the accuracy of ASR

The distribution of confidence suggests that even with custom language models, the accuracy of ASR needs to be improved so that students can be more confident in the tool.

Since ASR is still not accurate enough to provide to DHH students as a primary and sole access service, we asked participants questions that expanded our understanding of their preferences for accuracy.

Two similar questions asked them to think about how accurate ASR was in their biology or statistics course—but without specifying an actual number. One question asked participants if they would accept that same ASR accuracy rate in all of their classes in the future if they used only ASR with no other access services. Figure 4 depicts the number of participants who would accept or not accept the current accuracy rate of ASR *without* other access services.

One participant did not answer this question, possibly because this question was written almost exactly like a similar question.



Figure 4. Acceptance of ASR without other access services

A similar question asked if they would accept that same accuracy rate in all of their classes in the future if they used ASR **in combination with** other access services. Figure 5 depicts the number of participants who would accept or not accept the current accuracy rate of ASR with other access services.

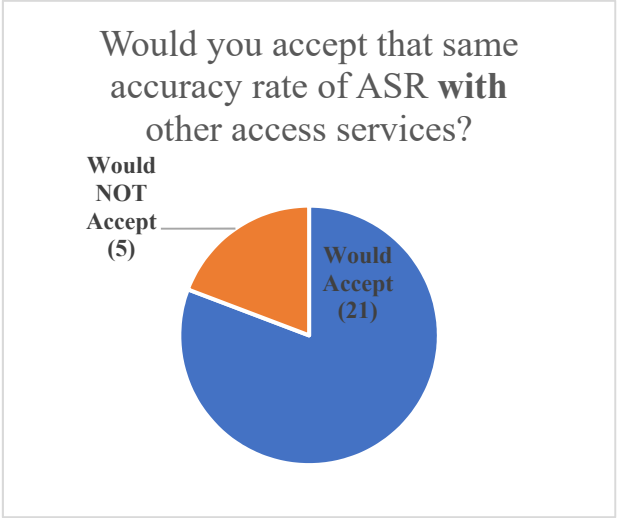


Figure 5. Acceptance of ASR with other access services

We asked them these questions about whether they would accept ASR with or without other access services because we aimed to understand the viability of ASR as a primary access service. The participants' responses suggest that students would be more likely to accept the current levels of accuracy of ASR **in combination** with other access services. Moving forward, we need to improve the accuracy of ASR in order to strengthen students' confidence in using ASR as a secondary and a primary access service.

4. ANALYSIS OF INTERVIEWS

To explore students' perspectives in deeper detail, we recruited 8 of the questionnaire participants for one follow-up interview each. They brought a range of experiences that enabled our understanding of how DHH students respond to ASR.

- 5 identified as female (P2, P3, P6, P7, P8) and 3 identified as male (P1, P4, P5).
- 4 signed in their interviews (P1, P2, P5, P6) and 4 spoke in their interviews (P3, P4, P7, P8).
- 4 took Introduction to Statistics II (P2, P4, P5, P6) and 4 took General Biology (P1, P3, P7, P8) in the previous semester.

The interview participants were asked four sets of questions that connected to the overriding research questions for this study. We comprehensively assessed and compared all participants' responses to each question. In the following subsections, we discuss our analyses of their responses and the themes that emerged.

4.1 Benefits and Limitations of ASR

4.1.1 ASR Provides Access but Is Inaccurate

Participants strongly indicated that the most beneficial aspect of ASR is how ASR provides students with access to the exact words that the instructor says. Students appreciate being able to look at the transcript when they miss what the instructor says or what the interpreter signs (P7), both in real time and for later reference. Students can see exactly what the instructor said and have "access to the words almost in real time," which allows them to "watch the presentation and the professor and the material all at the same time" (P5). The timing of ASR supported students because ASR is faster in timing than C-Print (P3) and students can watch the captions while the instructor talks (P4).

One participant explained that when she missed what the interpreter said, she could read the ASR transcript and use it "for support and to reference" what she missed (P2). Another participant appreciated being able to see what the instructor said when she did not hear something in class and being able to save the transcript to review and study later (P8). Their responses show how they used ASR as a supplemental support service during and after class.

Two participants decided that the most beneficial aspect of ASR was how ASR appears on the same location on screen as class material so that students do not have to look at a different device for captions (P1), and how ASR appears on students' laptops so students can read the transcript on their laptop while working (P6).

When asked about the least beneficial aspects of ASR, 7 of the 8 participants pointed to the inaccuracy of ASR as its least beneficial aspect while one participant seemed to misunderstand the question. One participant summarized their perspectives when explaining:

> "One of the disadvantages of ASR is its readability. And errors are—there were words that weren't accurate, and there would be a lag at times, and it's kind of hard to read." (P1)

Similarly, other participants critiqued how ASR made errors that "caused confusion" (P2), how it "doesn't pick up every word and it censored some words too [when thinking that the instructor said a curse word]" (P8), and how the "grammatical structure of the transcript was inaccurate" since "it didn't always get the words accurately and most times for words that had similar sounds, it would be written out or spelled differently" (P5). Their criticisms of ASR seem to focus on the readability of ASR—particularly how the errors or imperfections in spelling, vocabulary, and grammar hinders their ability to clearly read the captions.

4.1.2 Steady Reactions Throughout the Semester

We were interested in understanding any possible changes in participants' reactions to ASR throughout the previous semester as they became accustomed to seeing ASR in the classroom. During the interviews, participants were asked what their reaction to ASR was on the first day of class and 7 of the 8 participants used positive terms to describe their initial reaction to having an additional access service in class while some also observed the errors that ASR made. P5 remembered being "interested to see how it was going to perform" and thinking that "this was an extra tool to get access to all the information as possible" since he would be able to use ASR "during class to actually follow exactly what the professor was saying and also as a reference point." P8 explained that she is hard of hearing and did not request captioning services, so she was "pleasantly surprised" that ASR was there and that she could benefit from it without taking services away from other students. P2 mentioned that she had "a positive outlook on the first day of class" even though she knew "it had some inaccuracies."

One participant who did not use positive terms to describe her initial reaction to ASR on the first day of her statistics class in the previous semester had already taken a biology class with ASR in the past. P6 remembered how "in Biology it made too many mistakes" so she "wasn't really too excited about it in Statistics." Their responses show that students can be intrigued by the availability of a new access service in class but also be wary about the inaccuracy of the tool.

As the semester progressed with ASR in class, some of the participants' views of ASR evolved. When asked if and how their reactions to ASR changed throughout the semester, three participants described how they became accustomed to ASR in positive ways. One stated:

> "As we moved on into the semester, I realized the benefit of being able to access the content in the captioning in one place. So, it reduced the amount of information that I miss. So, then I started to feel like it really helped a lot." (P1)

Other participants' perspectives did not significantly change throughout the semester. P3 suggested that while her reaction "mostly stayed the same throughout the semester," the terminology became more complicated and so ASR "wasn't getting quite as accurate." P7 observed that the excitement wore off for her as she "noticed its limitations." She remarked that she knew that ASR was "a tool that I can use but not necessarily needed for everything." Their responses reflect the fact that ASR was provided as a supplemental tool, not a primary access service, for these students.

The participants' relatively steady reactions to ASR throughout the semester reveal that they continued to be aware of the positive and negative aspects of ASR.

4.2 Understanding the Class with ASR

4.2.1 Using ASR to Support Other Sources of Information

These DHH students seemed to use ASR to support their understanding of what happened in class as opposed to using ASR as their primary source. One participant explained:

> "I used it a lot for referencing after the class or the lecture, so most of the time I was watching the sign language interpreter and the instructor. If I missed something, I glanced down at the ASR transcript." (P2)

Her response shows how she used ASR transcripts when reviewing notes after class.

Participants were asked how they attended to different sources of information in the classroom, including the instructor, ASR, and other access services. Their varying responses reveal that there is no one way for any individual or group of individuals to sequence their attention in class. For instance, some participants used ASR as their primary source of information during specific moments in class but used ASR as backup to catch what they missed during other moments in class. Since ASR is used in a variety of ways to support students' learning, it is important that ASR be as accurate as possible to work across these contexts.

Students and instructors can also work together to improve how ASR is used in each classroom. P5 explained how he advocated for himself by asking his professor to say what she would write on the board before writing it down. As a result, "the professor adjusted her speaking style," which allowed him to have "access to what was going to be written." That allowed him to "look at the ASR, then the interpreter, and [then] look at the information." Collaboration makes ASR even more effective for a variety of learners.

4.2.2 Understanding the Instructor through ASR

7 of the 8 participants stated affirmatively that ASR helped them understand the instructor while the eighth (P6) did not have an opinion. P1 described how ASR helped him "understand the instructor a little bit better because it transcribed exactly what the instructor was saying." P2 acknowledged that it helped her understand the instructor, "but there were times [that] it made mistakes, and it was confusing."

The participants who spoke in their interviews revealed some insight into how ASR can help reinforce students' listening (P3, P4, P7, and P8). P3 explained that ASR helped her when she "would mishear things and it wouldn't sense and then I could go back to the ASR and see what she was actually saying." P7 described how missing the little things would "eventually add up," so it was helpful to be able to read what she missed. However, ASR has its limitations in helping students understand the instructor. P8 observed that her instructor generally spoke clearly. She explained, "sometimes if I didn't understand her [the instructor], the [ASR] captioning didn't get the right word either, it was something wrong. But it did help to some degree." It seems important then, that ASR be improved to ensure that students who listen to the instructor are able to understand all pieces of information so that minor errors do not accumulate.

4.2.3 Reading ASR in Class Is Not Necessarily Straining

The participants' responses suggest that when ASR is accurate, watching ASR can sometimes be less straining for students than viewing other access services. Three participants who signed in their interviews explicitly mentioned how watching ASR was less straining than watching the interpreter or than going back and forth between the interpreter and ASR (P1, P2, P6). P4 and P8 compared reading ASR with watching captions on television. These participants' responses show how processing information in different modalities can be tiring for DHH students in mainstream settings, which makes it even more important to make sure that class material is conveyed as accurately as possible to students.

Several participants mentioned how specific aspects of ASR could be fatiguing. P7 explained how it was frustrating because "there's the delay mostly and then also you have to watch every word written typed out, so that was kind of straining." One participant explained:

> "It was very, very broken up and they have very short segmented pieces and spit those out into separate lines and sometimes that would be a little bit distracting because it would be take 5 or 6 or 7 lines or bubbles paragraph whatever to finish a single sentence from the professor. So that did make it a little bit harder to understand but at the same time, it was much easier than just a solid dense block of text." (P3)

Their responses suggest that while the actual process of reading ASR is not necessarily straining, the visual appearance of captions affected their ability to efficiently read material on screen in the classroom.

4.3 Confidence in ASR

4.3.1 Relative Confidence in ASR

When explicitly asked about their confidence in ASR in their biology or statistics class, 6 of the 8 participants expressed confidence in ASR. Their level of confident may reflect how they used ASR to support their understanding of what was happening in class, not as their primary means of accessing information. One participant explained:

> "I felt confident in the ASR because I thought it was helpful. When I used it, after class, I could save the transcript and reference it later…. It became kind of my notes for class." (P2)

This participant's response shows that she was confident enough in ASR to use it as reference notes after class. Two of the participants who signed did not fully trust ASR:

> "I felt like watching the ASR I didn't always trust it. I would not look at individual words. I just look at the context of what was being said and try to understand everything that help me understand the big picture not the details. For the big picture, I used ASR. For the details, I trust the interpreter." (P1)

> "I don't really trust ASR. I know it's not a hundred percent accurate and sometimes if the word DEAF comes up it would spell D, E, A, T, H for death.… But I still have enough content to understand the concept." (P6)

Their responses show that, while they did not fully trust ASR because of the mistakes that ASR made, they still used ASR to support their understanding of class.

The other participants explained why they were confident in ASR. P3 trusted it after comparing ASR with TypeWell captioning to make sure they were similar. P4 explained: "I remember the first time I saw ASR in STAT 1, it's with the same teacher–ASR wasn't really that good, but when I saw […] it improved in STAT 2, I began to trust it." P5 trusted ASR because it provided access to specific vocabulary.

Two of the participants who spoke in their interviews explained how they had partial trust in ASR.

> "I was fairly confident. Because if [it] got it wrong I could look at the letters and can figure out what general area context was…. [Not very often] when there was a long delay, then all the words were jumbled, I was like I can't trust this." (P7)

> "I was pretty confident with it. A lot of the mistakes were small… It knew a lot of the words she was saying so even the big words about biology that aren't commonly used every day—it picked up on those words. And I like that about it. I think she mentioned once that like ASR had to learn the biology words.… I think that helps a lot with how accurate it was." (P8)

Overall, their responses show that the error rate in ASR may have influenced students' abilities to trust ASR—and that they used contextual information and other resources (including interpreters) to help them understand what was happening despite the errors that ASR made.

In addition, the 8 participants seemed to show agreement in stating that they would like ASR in their other classes while acknowledging that it can be difficult for ASR to accurately transcribe some content and clarifying that ASR would only work with certain class materials. As an example, P6 expressed uncertainty about using ASR in English class because of the lack of clear punctuation in ASR generated captions. Customizing ASR for different classes could strengthen the accuracy rates and increase students' confidence in ASR in different classes.

4.3.2 Comparison of Real-Time Captioning Services

Participants noted the differences between ASR and other real-time captioning services (C-Print and TypeWell), particularly that ASR can stay on pace with the instructor and allow them to get the necessary information faster. While several participants observed that human captionists do not always transcribe everything and can fall behind (P2, P6, P8), other participants pointed out that human captionists can correct their mistakes (P1, P3, P5) and can format captions to make captions easier for students to read (P3). P4 expressed a preference for ASR and commented: "I think out in the real world I'd be using ASR more than C-Print." His comment reveals how ASR can provide DHH individuals with direct and timely access to spoken information in and beyond the classroom.

These participants' responses may be influenced by the fact that students are generally not responsible for managing the costs of access services in the classroom. The relatively low cost of ASR compared to other access services may be a factor that leads to increased implementation of ASR throughout different institutions. While developing ASR as an access service, we recognize that ASR must be sufficiently accurate and readable for students to be able to be confident in the service.

4.4 Accuracy and Latency/Timing of ASR

4.4.1 The Need for Accuracy and Timing

Finally, we asked participants about their preferences for the accuracy and latency of ASR because improving one factor can worsen the other. Interrogating the relationship between accuracy and latency helps identify students' tolerance for these aspects, which in turn can inform further research on improving the accuracy and latency of ASR.

While students would prefer for ASR to be both accurate *and* timely, if they were forced to choose between the two, three of the participants would choose speed (P1, P2, and P5) and five of the participants would choose accuracy. One participant chose accuracy while insisting that ASR also be "instantaneous" and saying, "as long as the ASR keep up a good speed with how the person is talking, I'm OK with that"

(P6). Participants explained why speed was important to them. P2 explained that she used ASR to participate in class: "If there's too much of a delay, I'm not able to participate." Similarly, P5 explained that he would choose speed over accuracy in his statistics course because the instructor talked fast and he needed ASR to be timely enough in order to answer the questions that the instructor asked. In other cases, however, it would depend on the class and he would prefer more accuracy.

The other participants expressed a preference for accuracy over timing as long as the delay was not extensive. Overall, their responses show that students prefer for accurate *and* timely ASR in order to be able to access the learning environment.

Two participants directly mentioned that the delay would be relative to the speed of the instructor:

> "It depends on how fast the professor taught. If they talk really fast and ASR is lot slower, I feel like it be easy to miss something. Especially if they ask a question and you don't have like time to reply because you have to wait the four seconds and then like he's already moved [on]." (P8)

Their awareness of how instructors speak at different speeds shows that it is important for students to be able to stay on pace with the instructor through accurate and timely captions.

4.5 Responses to Sample Passages

To better understand how students respond to reading ASR generated captions with different accuracy rates, we asked interview participants to read 12 versions of passages generated by ASR engines and to declare whether or not they would accept each passage. Internally, we continue to test the accuracy rate of different ASR engines, but this study focuses on DHH participants' responses to the automatically generated captions created by different engines. Their responses show that students have concerns with more than the accuracy rate and need improved readability in captions that are designed to provide access to course information.

4.5.1 Preparing Passages and WER

To prepare for this stage of each interview, we started with an audio recording from one General Biology lecture. In order to obtain a completely accurate record of what the instructor said, we sent the audio recording to a captioning agency to be transcribed. After receiving the transcript from the captioning agency, a member of our team listened to the audio recording and proofread the transcript to verify its accuracy. This transcript served as our reference transcript. We then uploaded the original audio file from the class lecture to three ASR engines (Microsoft Custom Speech Service, Amazon, and IBM). The transcripts generated by these ASR engines became our hypothesis transcripts.

Prior to comparing the reference and the hypothesis transcripts, we removed punctuations, capitalizations, formatting, and extra spacing so that we could run the clean transcripts through SCLITE, a Word Error Rate (WER) calculator. After calculating the initial WERs for each transcript, we normalized the data by finding the differences that were marked as errors and comparing them with the original reference transcript. As an example, if the reference transcript displayed "it's going to want fifty" but a hypothesis transcript displayed "it is going to want 50," the initial WER would indicate that "it is" and "50" are incorrect. After identifying that both readouts had the same meaning, we adjusted the corrections in the transcripts and reran the hypothesis transcripts through SCLITE to calculate a new WER that we used as the indicator for the accuracy levels of each ASR transcript.

We selected four short passages from the reference transcript and selected the corresponding passages from the three ASR generated transcripts produced by Microsoft Custom Speech Service, Amazon, and IBM. The 12 passages produced by the engines had varying WERs that ranged from 5.7% to 29.4%. For the purposes of this research, we considered a passage with a 5.7% WER to be 94.3% accurate.

During the interviews, the interviewer showed each participant four sets of four passages. Each set included the reference passage and three versions of the same moment from the class lecture generated by ASR engines. The interviewer did not indicate the WERs of the passages. Each time he provided participants with a group of four passages, he asked them to read the sample passages produced by the ASR engines and declare whether they would accept each version within the context of their classroom environment.

The following bulleted list below and Figure 6 show the number of ASR samples that fell within each accuracy range and the percentage of participants who accepted passages in each accuracy range. As indicated in the following bulleted list and Figure 6, the acceptance rate for passages increased as the WER decreased.

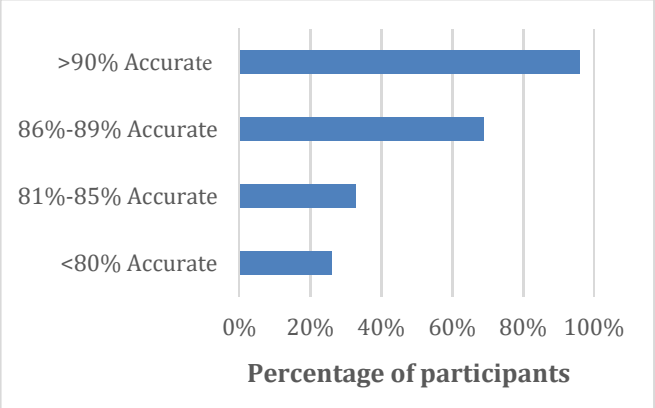


Figure 6. Percentage of participants who accepted ASR generated passages in each accuracy range

- **Less than 80% Accuracy / >20% WER (4 Samples):** 28% of participants accepted passages in this range
- **80–84% Accuracy / 16–20% WER (3 Samples):** 33% of participants accepted passages in this range
- **85–89% Accuracy / 11–15% WER (2 Samples):** 69% of participants accepted passages in this range
- **More than 90% Accuracy / <10% WER (3 Samples):** 96% of participants accepted passages in this range

The interviewer asked participants to justify their decision for accepting or not accepting each passage. Their responses provide additional clarification to why some students may accept passages with varying WERs.

4.5.2 Participants' Responses to Passages

The passage with the lowest WER in our study (Passage 1, Sample C) was produced by Microsoft Custom Speech Service with a custom language model and had a WER of 5.7%. All 8 participants accepted this passage. Three out of 8 participants commented that one mistake was "a sexual production" instead of "asexual reproduction." Even with that minor error, one said, "I could still understand the context" (P1), and another said that because "most of the keywords are there and the information is accurate, I would accept it" (P5). Their comments indicate that a few errors can be accepted when most of the contextual information and keywords are present.

In contrast, another version in the same group (Passage 1, Sample A) was produced by Amazon and had a WER of 19.5%. That engine made errors when transcribing "mitosis," "meiosis," and "cells" by displaying these terms as "my toast is," "my hostess," and "tells." Only one participant accepted this passage. The other participants said that these mistakes were confusing and that they could not understand the words. One stated, "a big problem with this would be the vocab words…. It would be hard to figure out what that meant– 'mitosis' or how to spell it" (P8). Another mentioned, "usually when the ASR messes up, you can read it and think about what it's trying to say, but this passage is too confusing" (P7). Their responses show how students have to actively attempt to decipher captions when errors appear.

When responding to other versions, participants explained how specific word errors made it hard for them to read the passages. For instance, a passage with a WER that was greater than 20% (Passage 4, Sample A) included the following errors: "ends up with a fork in our I" as well as "if you have 3 alarm well to" instead of "if you have three of them, two get silenced." 6 of the participants rejected the passage with one stating, "it's difficult to understand what's trying to be said" (P1). Two participants said they accepted the mistakes (P4, P7), with one explaining, "it gets most things right and the other mistakes I will have to go through it again and read it in order to understand" (P7). These two participants spoke in their interviews, which might suggest that students who are comfortable with spoken English might be more willing to resolve errors based on sound, such as "I" and "eye." To make ASR more readable for all students, researchers should continue to improve the accuracy rate and minimize the amount of effort that students need to exert as they try to figure out the actual words that have been said.

While participants were less likely to accept the passages with lower accuracy rates, participants accepted some passages with low accuracy rates. In those cases, the errors were relatively minor and did not have a large impact on the readability of the passages, which allowed participants to accept these passages.

5. IMPLICATIONS FOR ASR IN THE CLASSROOM

DHH participants' responses in this study informed our understanding of how to improve ASR as an access service for students in mainstream settings. Their responses reflect how ASR was provided as a supplemental tool, not a primary access service, for these students. Other students who directly rely on ASR as their primary means of access may be less likely to accept errors and may require additional improvements in ASR. We provide a set of implications for researchers who are considering designing and embedding ASR in other courses:

1. **ASR Benefits DHH Students When Supplementing Other Access Services.** Our primary finding is that participants perceived ASR as a helpful tool that enabled them to access spoken information throughout class. Some used ASR to catch what they may have missed when listening to the instructor or watching the sign language interpreter. Crucially, our custom language models made it possible for us to meet a minimum accuracy rate required for maintaining our ethical standards of providing access services that benefit students in the classroom. We are continuing to study and improve our custom language models and will disseminate our findings in the future.

2. **Improve the Accuracy and Speed of ASR Prior to Providing ASR as a Primary Access Service**. Participants' responses show that students need to be able to read and fully understand the words that are spoken in real time. While increasing the quality of ASR generated captions may cause a tradeoff in latency [9], improving accuracy *and* speed could increase students' confidence in using ASR as a primary access service. Slightly inaccurate ASR might be more acceptable as a supplementary access service for a class in which other access services are provided. Before providing ASR as the sole access service for a class, researchers should improve the accuracy rate of ASR to ensure that the engine meets or exceeds the accuracy of other real-time captioning services. This aligns with research (as noted in Section 2.1) on the importance of accuracy for real-time captioning.

3. **Place ASR Generated Captions Near Sources of Information:** Participants showed a preference for the placement of ASR generated captions near course slides. Participants who signed also described how watching the interpreter can be straining, so ASR generated captions should be placed close to other sources of information,

including the instructor and interpreter. To supplement research on the placement of ASR [18], more work should continue on minimizing the viewing distance between different sources of information in the classroom in order to minimize student fatigue.

4. **Work with Students and Instructors to Improve How ASR is Read:** Some participants described how they became accustomed to using ASR in class and mentioned the importance of instructors speaking at a pace that enables ASR to keep up. Researchers could collaborate with DHH students and instructors to improve the flow of speech that is transcribed into captions that are read in the classroom.

These points remind us that the presence of a custom language model linked to an ASR engine may not be sufficient in making mainstream settings fully accessible. In addition to feeding ASR systems with key vocabulary words and other course-specific concepts, we need to continue to improve the readability of ASR so that students can be confident in ASR as an access service.

6. LIMITATIONS AND FUTURE WORK

This research study was primarily limited by the small sample size for the questionnaire, which did not allow for conclusive statistical findings about correlations between students' experiences (with access, communication, and hearing levels) and their experiences with ASR in the classroom. In addition, participants were asked to recall their experiences in their course from the previous semester and may not have been able to remember all details; yet, we attempted to mitigate this limitation by collecting their comprehensive responses early in the following semester.

A limitation of our analysis of participants' responses to passages is that participants read static transcripts of passages as opposed to reading the flow of captions generated in real time. Their interpretations of the readability of captions in real time will differ, and future research projects should analyze students' responses in active learning settings.

Our findings provide guidelines for future studies that could expand this line of research by collecting more information from a larger sample of students in a larger range of courses. Researchers can examine how to improve the use of ASR in STEM courses and other disciplines. All in all, researchers need to continue to improve the accuracy and overall readability of ASR so that diverse learners can confidently and directly access spoken course material in real time.

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